

Classification using Naive Bayes in WEKA

Specifications Required

- WEKA software installed
- Dataset in .arff format (e.g., weather.arff, iris.arff)
- Basic understanding of classification and Naive Bayes algorithm

Aim

To classify data using the Naive Bayes algorithm and understand the classification rule process in WEKA.

Theory

Naive Bayes is a probabilistic classifier based on Bayes' theorem with an assumption of independence among features.

Bayes' Theorem:

$$P(C|X) = (P(X|C) * P(C)) / P(X)$$

Where:

- $P(C|X)$: Posterior probability
- $P(X|C)$: Likelihood
- $P(C)$: Class prior
- $P(X)$: Predictor prior

Steps to Perform in WEKA

1. Open WEKA GUI Chooser and select Explorer.
2. Load a dataset (e.g., weather.arff) by clicking on 'Open file...'.
3. Go to the 'Classify' tab.

4. Click 'Choose' -> bayes -> NaiveBayes.
5. Set test option (default: Use training set).
6. Click 'Start' to run the classifier.
7. Observe the results: accuracy, confusion matrix, class probabilities.

Conclusion

The Naive Bayes classifier in WEKA provides an effective way to classify data using probability-based rules. It is easy to apply and interpret, especially useful when attributes are independent.